Lab 1 Activity Sheet

BEE 4750/5750, Fall 2026

Name: _____

NetID: _____

This sheet covers the written and derivation parts of Lab 1 (Convergence and Discretization). Do the coding in `lab01.ipynb` or `lab01.jl`; fill out this sheet, and hand it in at the end of class (or you may scan it and submit by 9pm if you want a little more time to work). See `lab01.ipynb` or `lab01.pdf` for the full model writeup, including parameter values, and problem statements.

Problem 1: Discretize by Hand

Starting from

$$\frac{dG}{dt} = k\left(L_0\, W(t) - G(t)\right),$$

derive the forward Euler update rule for $G(t + \Delta t)$ in terms of $G(t)$, Δt, and the other model quantities. Show your work.

Name: _____

Problem 2: How Small Does Δt Need to Be?

Problem 2a: Characteristic Timescale

Compute the characteristic timescale, $1/|\text{rate}|$, for this model. Use it to propose a starting Δt that should be "well inside" it, and justify your choice in 1-2 sentences. Do this *before* running any code.

Problem 2b: Convergence Results

Simulate your model and try to find a step size Δt for which the peak CH_4 rate converges. After conducting your convergence study, write down your results in the table below.

Δt [yr]	Peak G [m^3 CH_4/yr]	Error vs. reference	Error Improvement

Make a log-log plot based on these values. In 1-2 sentences, connect the error you observe to the argument from class about forward Euler error based on Taylor series.

Problem 3: Make and Justify a Recommendation

Suppose you need to recommend a step size to use for sizing a flare system from this model. In 2-4 sentences, state the Δt you'd actually use and justify it, referencing **both** the characteristic-timescale bound from Problem 2a (a

floor from theory) and your empirical convergence check from Problem 2b (a stopping rule from data) — not just "smaller is better."